

Charan Lingamgunta

STAFF SOFTWARE ENGINEER

AI/ML PLATFORM | LLM INFRASTRUCTURE | DISTRIBUTED SYSTEMS

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PROFESSIONAL SUMMARY

Staff Software Engineer with 15 years building AI/ML platforms and high-scale distributed systems for regulated healthcare, financial services, and government clients. Architected the agent execution, LLM fine-tuning, and developer platforms behind a commercial enterprise AI suite: 20+ production fine-tuned models, 2,500+ governed agent tools, and 200+ engineers served, on a gateway foundation processing 10B+ transactions daily.

CORE SKILLS

AI & ML Platforms: LLM Fine-Tuning (PEFT, LoRA), Evaluation Pipelines (MLflow GenAI eval, LLM-as-Judge), LLM Observability & Tracing (MLflow, OpenTelemetry), Model Serving (vLLM), Agent Harness & Platforms, RAG (hybrid BM25 + kNN, RRF), MCP, DSPy, Guardrails, Hugging Face, PyTorch, CUDA, Ray

Platform & Distributed Systems: Distributed Systems, Event-Driven Architecture, Pub/Sub & Messaging (JMS, AMQP, Kafka), API Gateways, Developer Platforms, Multi-Tenant Systems, High-Availability Design

Cloud & Infrastructure: AWS (EC2, S3, RDS, ECS, Lambda, CloudWatch), GCP, Docker, Terraform, OpenSearch, Redis, Airflow, Spark, SkyPilot, Jenkins, GitHub Actions

Languages: Python, Java, TypeScript, JavaScript, C#

Frameworks & Developer Tooling: Backstage, React, Node.js, FastAPI, Django

Security & Compliance: Sandboxed Code Execution, PII Detection & Redaction, Safety Guardrails, OAuth 2.0, SAML, LDAP, HSM, HIPAA, HITRUST, PCI-DSS, SOC 2, FIPS 140-2

PROFESSIONAL EXPERIENCE

Staff Software Engineer | Forum Systems | Needham, MA

Jan 2022 - Present

Promoted to staff-level ownership across LLM platform, agent execution, and developer platform initiatives, shaping shared architecture, evaluation and security standards, and ADR practices across teams.

AI SecureExec, Agent and Tool Execution Platform

Feb 2023 - Present

- Built the enterprise agent harness (DSPy) behind a commercial AI platform: analyst and data science teams run 10,000+ governed agent invocations daily through ReAct and CodeAct workflows, each agent bound to its own model, with results streamed over SSE or dispatched to background workers.
- Designed per-tenant sandboxed execution for user-authored agent code: an AST-gated interpreter enforcing import allowlists, secrets scanning, and runtime limits at parse time and on every execution step, so semi-trusted analyst code reaches a live multi-tenant runtime without a platform engineer reviewing every change.
- Published an MCP-compatible tool registry with discovery, scoped per-identity authorization, and SOC-aligned audit, giving teams a governed path to reuse 2,500+ enterprise tools in agent workflows; both DevX and SecureExec expose MCP servers.
- Designed agentic RAG pipelines over OpenSearch hybrid retrieval (BM25 + kNN with reciprocal rank fusion and query decomposition), with judge-in-the-loop correction that catches both hallucinations and false gaps; self-hosted embeddings keep data in-tenant and eliminate per-token cost at 50,000+ documents ingested monthly.
- Built the offline evaluation gate on MLflow: a versioned golden dataset over the regulated corpus, scored by trace-aware LLM-as-judge graders that assess tool calls and reasoning steps, not just final answers, run in CI so prompt, model, or tool changes that regress faithfulness are blocked before release.
- Instrumented MLflow tracing across agent execution, tool usage, and policy decisions, with OpenSearch structured logging, giving teams replayable traces to debug and measure agent behavior in production.

- Implemented a multi-provider LLM gateway (LiteLLM) routing per-tenant traffic across self-hosted, fine-tuned, and Bedrock-hosted models with per-tenant access policies, context-window-aware token budgeting, and per-call cost emission, letting regulated tenants pin to private models for data residency.
- Integrated agent inputs and outputs with the enterprise PII detection and redaction control plane, enforcing content-moderation and safety guardrails in HIPAA-regulated deployments.

DevX Fine-Tuning, LLM Fine-Tuning and Deployment Platform

Jan 2023 - Jun 2025

- Developed the LLM fine-tuning platform behind 20+ production models at enterprise clients: one workflow to onboard Hugging Face models, launch PEFT/LoRA jobs, and compare runs in MLflow, with LLM-as-judge plus human review gating every release.
- Lowered training cost with SkyPilot spot orchestration and Ray distributed training across AWS and GCP; served models on GPU-backed vLLM with OpenSearch monitoring of training and production behavior.
- Built versioned dataset pipelines for Excel and JSON corpora with repeatable preprocessing and reproducible train/validation/test splits.

DevX, Developer Experience Platform

Jan 2022 - Present

- Led architecture for the developer platform serving Fortune 500 healthcare and financial services clients, unifying API discovery, deployment visibility, dependency mapping, and operational metadata; the largest single deployment serves 200+ engineers and 1,000+ APIs.
- Converted 1,000+ API specifications into 2,500+ governed, MCP-compatible tools through an OpenAPI-to-tool pipeline, removing the review-queue bottleneck for agent tool publication.
- Built the safe self-authoring path that drove enterprise adoption: analysts write custom tools in a Monaco-based editor against an allowed Python library set, validate them in sandbox tests, and publish only on passing, gated by entity-level RBAC and tenant isolation.
- Delivered multi-tenant capabilities for regulated deployments (client-specific plugins, Okta/GitHub/local identity integration, entity-level permissions) and mentored 5 engineers, 2 of whom were promoted to senior roles.

Senior Software Engineer, Distributed Systems | Forum Systems | Needham, MA

Jun 2014 - Dec 2021

Forum Sentry, Enterprise API Gateway and Builder

- Architected the 4-stage bidirectional streaming pipeline carrying 10B+ transactions daily across HTTP, SOAP, JMS, AMQP, SFTP, and S3, with zero-copy passthrough, automatic retry/failover, and 7 timing checkpoints for stage-level latency attribution.
- Cut backend load 40-70% with multi-tier API response caching (memory, disk, Redis; RFC 7234-compliant) with request coalescing for cache-stampede protection, configured through no-code cache policies.
- Designed the no-code API builder: 60+ processors for security, data transformation, routing, and validation, sandboxed inline JavaScript (on the JVM) and Python scripting, and 20+ partner, cloud, and database integrations, cutting client onboarding from weeks to days.
- Engineered active-passive failover with 9 load-balancing strategies, sub-minute recovery, and 99.99% uptime targets for financial and government workloads, on FIPS 140-2, HSM-backed cryptography.
- Implemented the authentication subsystem supporting 11 enterprise identity methods, with auth-side caching and connection pooling that eliminated 90%+ of directory roundtrips.

EARLIER EXPERIENCE

- **Software Developer Intern, Mimecast, Watertown, MA (2013):** enterprise email encryption application in C#/WPF with PowerShell test automation.
- **Associate Software Engineer, AppLabs (CSC), Hyderabad, India (2009 - 2012):** reporting and notification modules for an e-learning platform serving millions of users.

EDUCATION

Master of Science in Computer Information Systems, Northeastern University

Boston, MA

Bachelor of Engineering in Computer Science, JNT University

India